

# Research Talk

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It mainly focus on developing rigorous, computational efficient and generalizable Statistical Methodology to understand the change of data - specifically in the deployed Machine Learning(ML) systems.

Main Scientific & Practical Questions to address:

- Given two batches of data  $D_{exist} = (\mathbf{X}_{exist}, Y_{exist})$  and  $D_{new} = (\mathbf{X}_{new}, Y_{new})$ . How to know whether the difference is from concept drift( $P(Y|\mathbf{X})$ ) or covariate shift( $P(\mathbf{X})$ )?
- How to understand the change of data from the feature level?
- Given a deployed ML model in the production system, how to know when and how should we update it?

# Overview of Methods

## Overall Roadmap for Thesis: Testing, Understanding and Characterizing Distribution Shift

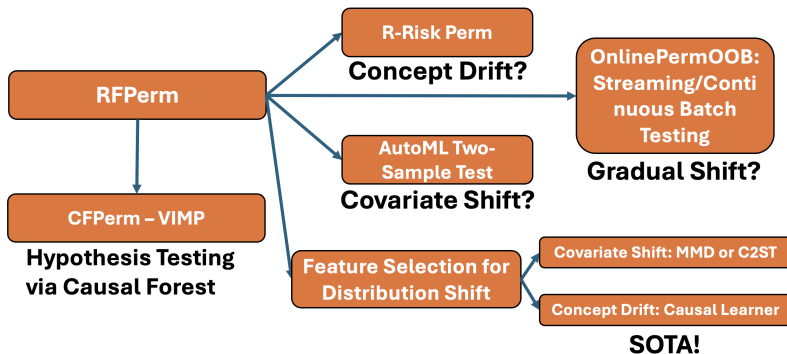


Figure: Overview of the Methodologies we have developed.

Permutation test to compare validation error(OOB Error) and test error(Prediction Error) in Random Forest.

- Random Forest, well known for its robustness across a wide variety of different scenarios, would be the best choice to isolate the effect of OOD(out-of-distribution).
- It would be capable of characterizing both **Concept Drift** and **Covariate Shift** – Density Aware nature of Random Forest.
- Generalizable to arbitrary Machine Learning Models on arbitrary Modality of data - as long as the validation set is well-defined.

Leverage Causal Inference Method to detect Distribution Shift - via Variable Importance.  $D_{exist}$  as Control Batch( $T = 0$ ),  $D_{new}$  as Treatment Batch( $T = 1$ ).

- Permute-then-refit the Causal Learner, looking at the variable importance(LOCO or permutation variable importance).
- Serves as a Feature Selection Technique for Distribution Shift.
- The variable importance in this framework itself can be interpreted as the quantification of the contribution to certain types of distribution shift, potentially OOD variable importance.

Leverage Causal Inference Method, R-learner to develop hypothesis testing procedure for isolating concept drift from covariate shift.

- The motivation of R-learner is similar to that in the Double/Debiased Machine Learning(DML) - Remove the effect of the covariates.
- Based on Projection Theory, the  $\tilde{Y} = Y - m(X) = (I - \Pi_X)(Y)$  and  $\tilde{T} = T - e(X) = (I - \Pi_X)T$ . The R-risk is invariant under Covariate Shift, isolating of the Concept Drift. The first ever Statistical Methodology that can detect **Concept Drift Only**.
- Can be generalizable to arbitrary Objective Functions in literatures of Causal Inference.

In production, to conduct feature-level attribution, the following aspects can be leveraged:

- For Covariate Shift only, MMD+LOCO(leave-one-covariate-out) or train a random forest to distinguish two datasets is efficient.
- For Concept Drift, leveraging the Causal Attribution procedure for Feature Selection.
- Proactive and Actionable diagnostic tools for feature quality deterioration.

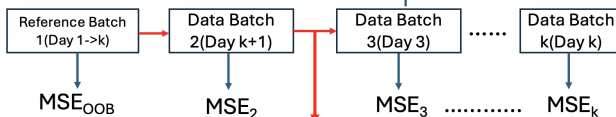
Suppose a deployed model  $\hat{f}$  is launched in Production. In the streaming data setting, new observations  $\cup_{i=1}^n \{\mathbf{X}_i, Y_i\}$  coming continuously, we aim to give people an idea of when the gradual distribution shift become significant - trigger model update.

- A procedure based on empirical p-values is proposed to assess the extremity of the current predictive error relative to those in the deployed model.
- An online False Discovery Rate (FDR) control procedure is incorporated to effectively mitigate false alarms.
- **Model Agnostic** as long as a valid predictive model is well-defined and near-real-time inference is enabled.

# Methodology 4: Illustration

When the distribution shift become significant for **already-deployed model**?

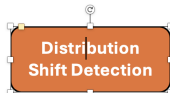
Random Forest as Reference Model



Empirical p-value

Prediction

$$P - value = (1 + \{\#MSE_{i,i=1\dots,k} < MSE_{ref}\}) / (1 + k)$$



Online Anomaly Detection

Any-time Valid Type-I Error Control

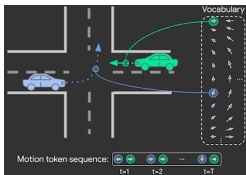
Flexible: Batch Setting/ Streaming Setting

- Very Robust under High-Dimensional Covariates
- Change-Point Detection via Monitoring Predictive Performance

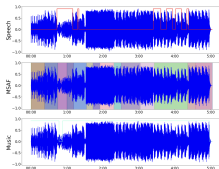
Figure: Streaming Testing for Gradual Distribution

# Methodology 4: Flexibility and Modality Agnostic

The methodology is model-agnostic, with demonstrated effectiveness across diverse data modalities—making it broadly applicable to any ML system without architectural modification.



(a) Motion Planning



(b) Audio



(c) Video Streaming

|                          | Train           |                        |               | Test                  |                         |
|--------------------------|-----------------|------------------------|---------------|-----------------------|-------------------------|
| Satellite Image (x)      |                 |                        |               |                       |                         |
| Year / Region (d)        | 2002 / Americas | 2009 / Africa          | 2012 / Europe | 2016 / Americas       | 2017 / Africa           |
| Building / Land Type (y) | shopping mall   | multi-unit residential | road bridge   | recreational facility | educational institution |

(d) Functional Map (Image)

# Feature Selection for Distribution Shift

- Given two batches of data, how should one know how the shift can be explained by the individual features? It can be formulated as a Feature Selection problem for the Distribution Shift(both covariate shift and concept drift).
- It's a problem that every data scientist will encounter with real-world scenarios, tractable practical values as well as the promising extensions build upon the current Statistical & Machine Learning Methodologies.
- This procedure can be embedded in the deployed ML systems - along with the existing practices including KL-divergence & PSI.

# Feature Selection for Covariate Shift

We want to understand the difference of  $P(\mathbf{X}_{exist})$  and  $P(\mathbf{X}_{new})$  - which subset of features are contributing significantly to the covariate shift?

- **MMD-LOCO**: Maximum Mean Discrepancy(MMD) equipped with the leave-one-covariate procedure - calculate the difference as the ranking mechanism for the feature selection.
- **RF-Domain Classifier**: A binary classifier is trained on distinguishing the two batches of datasets  $\mathbf{X}_{exist}$  and  $\mathbf{X}_{new}$ , borrowing the ideas from the C2ST(Classifier Two-Sample Testing Procedure) and regard the variable importance ranking for the feature selection criterion. The vanilla version is enough, won't need compound versions like **Boruta** or **ShadowVIMP**!
- The random forest based procedure is highly competitive in comparison to the existing SOTA - **FSL-Net**.

# Feature Selection for Concept Drift

- Regarding of concept drift, the causal inference methods can be leveraged including **permuCATE(concise feature selection)** & **causal forest split VIMP procedure(most parsimonious, relatively efficient)** are recommended in practice.
- The causal inference method is designed to detect the equivalence of conditional mean  $E[Y|X, T = 1]$ ,  $E[Y|X, T = 0]$ . The numerical studies and real-data analysis also demonstrated the SOTA performance of the causal inference methods in a wide variety of types of data.

Understanding distribution shift from feature level has direct practical value in the production ML System.

- The two-step procedure for the feature selection can enable the transformation from the hypothesis testing procedure to the feature selection mechanism
- The covariate shift & concept drift attribution procedure, coupled with other model monitoring practices, can enable the near-real-time, automatic feature removal & granular root-cause diagnosis with the reduced MTTD/MTTA.

## Integration into the Production System:

- The proposed feature selection procedure for distribution shift can be seamlessly integrated into existing model monitoring pipelines, complementing conventional metrics such as PSI, KL divergence, and the KS statistic in deployed ML systems.
- While fully assumption-less identification of distribution shift drivers is infeasible, the feature selection for distribution shift procedure is still capable of extracting a ranked list of the top-k features contributing significantly to the observed distribution shift, providing actionable references for Business, Operation and Engineering teams.

This thesis propose two lines of Statistical Methodologies: Streaming Testing and Batch Testing procedures that is highly generalizable to a wide variety of production scenarios

- Batch Testing: Parallelization available for Permutation Test, computationally feasible for conducting the test each 5 minutes.
- Production Landing for the Streaming Testing: Near real-time inference & hypothesis testing for the Streaming Observations to trigger Model Refresh Pre-Signals.
- Highly Flexible: all of these methodologies can be adapted to a wide variety of types of datasets & modalities.

# Future Directions

Several promising applied research avenues naturally extend our framework, each offering tangible value for production ML systems:

- **Streaming Monitoring & Detection:** Extend to continuous micro-batches using neural encoders (TarNet/DragonNet) for online streaming OOD detection.
- **Modality Generalization:** Generalize the feature selection to multi-modal settings (AV, motion planning) beyond tabular data.
- **Region Selection:** Develop localized sub-population selection strategies for OOD detection and domain adaptation characterization procedures with respect to different regions.
- **Model Monitoring + Contextual Bandit:** Several Deployed ML model, identify the robust feature subsets to ensure the model continuously generate business impacts - epsilon Greedy + Enhanced Feature Engineering is probably what you need!